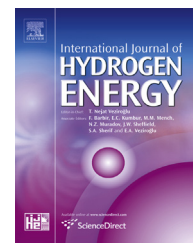


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Efficient hydrogen storage in up-scale metal hydride tanks as possible metal hydride compression agents equipped with aluminium extended surfaces

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ABSTRACT

In the current work, a three-dimensional computational study regarding coupled heat and mass transfer during both the hydrogenation and dehydrogenation process in up-scale cylindrical metal hydride reactors is presented, analysed and optimized. Three different heat management scenarios were examined at the degree to which they provide improved system performance. The three scenarios were: 1) plain embedded cooling/heating tubes, 2) transverse finned tubes and 3) longitudinal finned tubes. A detailed optimization study was presented leading to the selection of the optimized geometries. In addition, two different types of hydrides, LaNi₅ and an AB₂-type intermetallic were studied as possible candidate materials for using as the first stage alloys in a two-stage metal hydride hydrogen compression system. As extracted from the above results, it is clear that the case of using a vessel equipped with 16 longitudinal finned tubes is the most efficient way to enhance the hydrogenation kinetics when using both LaNi₅ and the AB₂-alloy as the hydride agents. When using LaNi₅ as the operating hydride the case of the vessel equipped with 60 embedded cooling tubes presents the same kinetic behaviour with the case of the vessel equipped with 12 longitudinal finned tubes, so in that way, by using extended surfaces to enhance the heat exchange can reduce the total number of tubes from 60 to 12. For the case of using the AB₂-type material as the operating hydride the performance of the extended surfaces is more dominant and effective compared to the case of using the embedded tubes, especially for the case of the longitudinal extended surfaces.

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Introduction

Hydrogen is a promising alternative energy carrier that could potentially facilitate the transition from fossil fuels to sources of green energy due to its high energy density (141 MJ/kg), variety of potential sources and low environmental impact [1]. Hydrogen can be stored as a high pressure gas or low temperature liquid or alternately, it can, by undergoing a chemical reaction, be stored in solid form metal hydrides or cryo-adsorbents [2,3]. Issues associated with hydrogen storage materials are the storage capacity, the thermal stability of the hydride, the kinetics of hydrogenation and dehydrogenation, thermophysical properties and crystal structures. Specifically fast reaction kinetics and high hydrogen storage capacity are the most important properties of the alloys used in thermal devices [4], compression systems [5–12] and heat pumps [13,14]. Generally the limiting factor controlling the hydrogen charging/discharging in a metal hydride reactor is the rate at which heat can be transferred between the hydride and the reactor. Due to the exothermic nature of the hydrogenation process, large amounts of heat are produced inside the reactor, forcing the equilibrium pressure to increase and subsequently lowering the driving force for the hydrogen storage. On the other hand, providing heat and consequently lowering the internal pressure is necessary for the dehydrogenation process to take place. The thermal management of a metal hydride reactor must take into account both the cooling during the hydrogenation process heating during the dehydrogenation process. Two active heat management techniques can be employed to accomplish this, internal heat management and external heat transfer [15,16]. Heat exchangers are used in many industrial applications the most widely used being those intended to exchange heat between two fluids. Less attention has been given to the design of heat exchangers involving the heat flow between a stationary reacting metal hydride powder and a cooling/heating fluid as required in the field of solid state hydrogen storage.

Despite many efforts dedicated to enhancing the rate of the external heat transfer, the performance of the metal hydride systems is still limited [17]. Therefore, internal heat transfer enhancements have become more attractive and can roughly be divided in two ways: 1) Enhancement of the thermal conductivity of the metal hydride powder by insertion of aluminium foam [18–20], copper wire net structure [21] or by creating metal hydride compacts [22–25], 2) Integration of heat exchangers inside the bed such as embedded heat exchanger tubes [26–28], finned tube heat exchangers [29–33], spiral heat exchangers [34,35] or heat pipes [36].

In a previous work [37], the authors described the simulation study of the operation of a two-stage metal hydride hydrogen compression system and studied the combination of different materials. The heat management process of the reactors for that study did not go any further than the fact that external heating/cooling jackets used for all the reactors, as the main focus of the study was the understanding the operation of the coupled compressor stages. In the current work, three different heat management scenarios are studied numerically using Finite Element Analysis software COMSOL Multiphysics 4.4 in order to decide which scenario, if

incorporated into an up-scale metal hydride tank, will result in the most efficient operation of the compression system. The first scenario involves the use of a differing number of plain cooling/heating tubes embedded in the reactor. The second scenario includes transverse Al surfaces extending from the cooling/heating tubes and the third scenario deals with longitudinal Al extended surfaces. For each scenario, a two dimensional study was conducted to optimise the number of cooling/heating tubes, diameter of the tubes (scenario 1), number of heating/cooling tubes, number of Al fins and fin thickness (scenarios 2 and 3). The optimised geometry was then modelled in 3D. In this study a novel AB₂-based intermetallic is introduced, validated and simulated under three different heat management scenarios for both the hydrogenation and dehydrogenation process. Furthermore, the results are compared with the performance of LaNi₅ in order to suggest a potential candidate material for the first stage of a multistage metal hydride hydrogen compression system. The results showed that the presence of the Al fins can reduce both the hydrogenation and dehydrogenation time and in terms of charging time, 60 cooling tubes may be replaced with only 12 or 16 finned tubes with faster kinetics. The effect on reactor performance was also compared using two different first stage storage materials. The first material was LaNi₅ and the second material was a Laves-phase AB₂ Ti–Zr based intermetallic.

Methodology

In the current numerical work the reactor performance was modelled by the simultaneous use of COMSOL Multiphysics chemical reactions, to implement the reaction of hydrogen with the metal hydride, and heat transfer in solids, for the heat transfer through the hydride bed and heat exchange components. Prior to conducting these simulations the performance of the model was validated using experimental results extracted from a lab-scale Sievert's type apparatus. After the validation process, the optimization study of the three scenarios was performed. For scenario 1, the impact of the number of cooling/heating tubes was explored by modelling the reactor performance using 18, 24, 30, 36, 48 and 60 heat exchange tubes. For scenarios 2 and 3, three different parameters were optimised i.e., the radius, number and thickness of the fins. For all three scenarios, the radius of the heating/cooling tubes was selected at 3.2 mm corresponding to a commercial available ¼ inch stainless steel tube. The expansion of packed beds during the hydrogenation process can produce additional stress on the vessel wall, therefore for the purpose of the model, the reactors are assumed to be filled to 50% at the beginning of the hydrogenation process. Table 1 presents all the details about the optimization parameters studied in the current model.

Model assumptions

A number of assumptions were made which serve to simplify the modelling process. These were:

- a) Initially the temperature and pressure profiles are uniform.

Table 1 – Parameters used for the optimization study scenarios 1, 2 and 3.

Scenario	Tube radius (mm)	Number of tubes	Number of fins	Fin radius/height (mm)	Fin thickness (mm)
1) Cooling/heating heat exchangers	3.2	18-24-30-36-48-60	–	–	–
2) Transverse finned tubes	3.2	8-12-16	20-25-30-35-40	5-6-7-8	1-1.5-2-2.5-3.5
3) Longitudinal finned tubes	3.2	8-12-16	6-12	2-3-3.5-4-5-5.5	0.15-0.25-0.35-0.5

- b) Thermal conductivity and specific heat of the hydrides are assumed to be constant during the compression cycle.
- c) The medium is in local thermal equilibrium which implies that there is no heat transfer between solid and gas phases.
- d) Hydrogen is treated as an ideal gas from a thermodynamic point of view.
- e) The bed void fraction remains constant and uniform throughout.
- f) The characteristics, e.g., the kinetics and thermal properties, of the bed are unaffected by the number of loading and unloading cycles i.e., bed aging is neglected.
- g) The bed fills the entire volume of the space between the fins and tubes.

Heat equation

Assuming thermal equilibrium between the hydride powder and hydrogen gas, a single heat equation is solved instead of separate equations for solid and gas phases:

$$(\rho \cdot Cp)_e \cdot \frac{\partial T}{\partial t} + (\rho_g \cdot Cp_g) \cdot \vec{v}_g \cdot \nabla T = \nabla \cdot (k_e \cdot \nabla T) + m \cdot \left(\left(\frac{\Delta H}{M_{H_2}} \right) - T \cdot (Cp_g - Cp_s) \right) \quad (1)$$

where, the effective heat capacity is given by;

$$(\rho \cdot Cp)_e = \varepsilon \cdot \rho_g \cdot Cp_g + (1 - \varepsilon) \cdot \rho_s \cdot Cp_s \quad (2)$$

and the effective thermal conductivity is given by;

$$k_e = \varepsilon \cdot k_g + (1 - \varepsilon) \cdot k_s \quad (3)$$

The terms ρ_g , Cp_g , Cp_s and m refer to the density of the gas phase, the heat capacity of the gas phase, the heat capacity of the solid phase and the kinetic term for the reaction respectively. The parameter that represents the void fraction is symbolised with ε . M_{H_2} represents the molecular mass of hydrogen and T represents the temperature.

Hydrogen mass balance

The equation that describes the diffusion of hydrogen mass inside the metal matrix is given by:

$$\varepsilon \cdot \frac{\partial(\rho_g)}{\partial t} + \text{div}(\rho_g \cdot \vec{v}_g) = \pm Q \quad (4)$$

where, (–) is for the hydrogenation process and (+) is for the dehydrogenation process, v_g is the velocity of gas during diffusion within the metal lattice (see [Momentum Equation](#) section) and Q is the so-called Mass Source term describing

the mass of hydrogen diffused per unit time and unit volume in the metal lattice.

Momentum equation

The velocity of a gas passing through a porous medium can be expressed by Darcy's law. By neglecting the gravitational effect, the equation which gives the velocity of gas inside the metal matrix is given by:

$$\vec{v}_g = -\frac{K}{\mu_g} \cdot \text{grad}(\vec{P}_g) \quad (5)$$

where K is the permeability of the solid and μ_g is the dynamic viscosity of gas and P_g is the pressure of gas within the metal matrix.

Kinetic expression

The reaction kinetic equation describes the expression for hydrogen mass hydrided and dehydrided per unit time and volume. The amount of hydrogen taken up is given by;

$$m_a = C_a \cdot \exp \left[-\frac{E_a}{R_g \cdot T} \right] \cdot \ln \left[\frac{p_g}{P_{eq}} \right] \cdot (\rho_{ss} - \rho_s) \quad (6)$$

The amount of hydrogen released from the hydride bed is given by;

$$m_d = C_d \cdot \exp \left[-\frac{E_d}{R_g \cdot T} \right] \cdot \left(\frac{P_{eq} - p_g}{P_{eq}} \right) \cdot \rho_s \quad (7)$$

where ρ_s and ρ_{ss} are the density of the hydride at any time and at saturation state respectively. C_{abs} and C_{des} refer to the pre-exponential constants for the hydrogenation and dehydrogenation process and the E_a and E_d are the activation energy for hydrogenation and dehydrogenation process respectively.

Equilibrium pressure

In order to incorporate the effect of hysteresis and the plateau slope (which is present in both stages of the compression cycle) for the calculation of the plateau pressure P_{eq} , the following equation was used [38,39]:

$$\ln \frac{P_{eq}}{10^5} = \left\{ \left[\frac{\Delta H}{RT} - \frac{\Delta S}{R} \right] + (\varphi_s \pm \varphi_0) \cdot \tan \left[\pi \cdot \left(\frac{x}{x_{sat}} - \frac{1}{2} \right) \right] \pm \frac{S}{2} \right\} \quad (8)$$

The plateau slope is given by the flatness factors φ_s and φ_0 and S represents the hysteresis effect which is $(\ln P_{abs}/P_{des})$ designated ‘+’ for hydrogenation and ‘–’ dehydrogenation. x and x_{sat} are the local hydride concentration at any given time and at saturation respectively. For the two studied materials

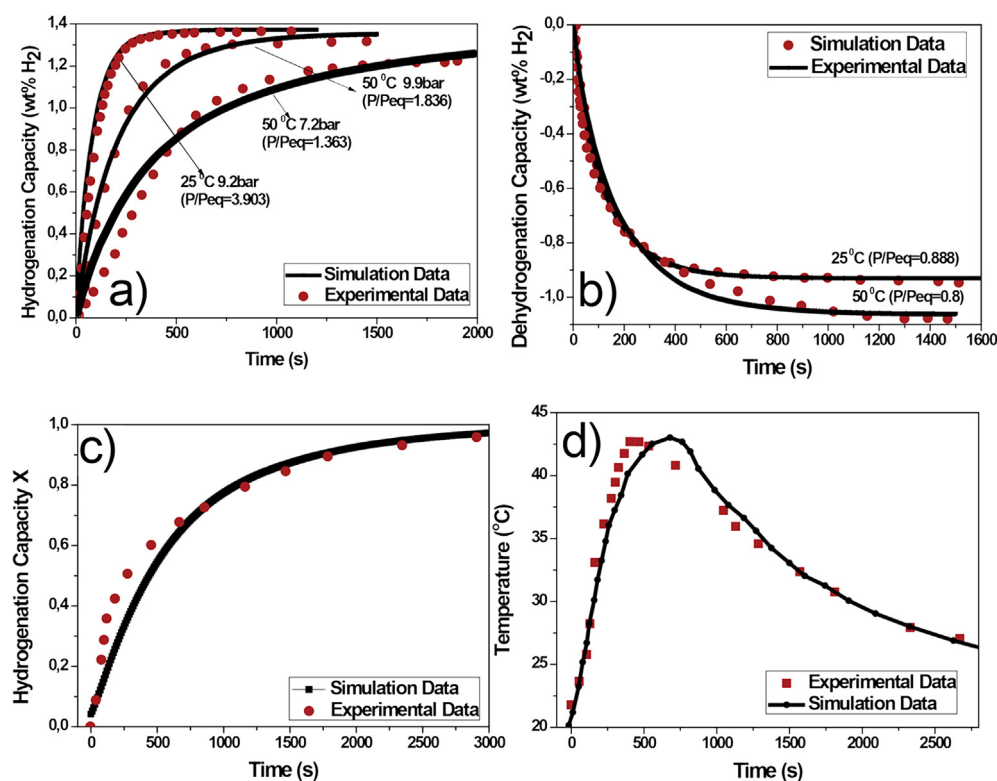


Fig. 1 – Validation of the predicted amount of hydrogen stored (a) for different temperatures and different pressures and released (b) for two different temperatures with experimental data extracted from a lab scale Sievert type apparatus. (c) shows the validation of the hydrogenation behaviour of the AB₂-intermetallic and (d) presents the validation of the temperature profile of the same material. The experimental results (red dots) with the simulation results (black lines) are in good agreement with a maximum deviation less than 5%. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

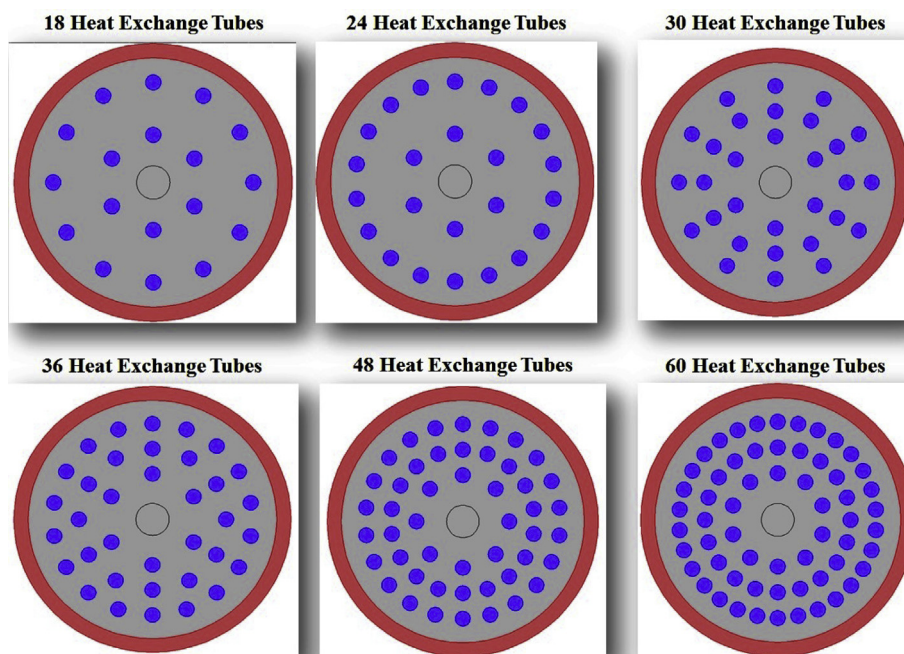


Fig. 2 – The geometries used for the study of scenario 1. Different arrangements of the co-central tubes were used and the different number of tubes used was 18, 24, 30, 36, 48 and 60. The central tube used for the hydrogen delivery during the hydrogenation process and for the hydrogen expulsion during the dehydrogenation process.

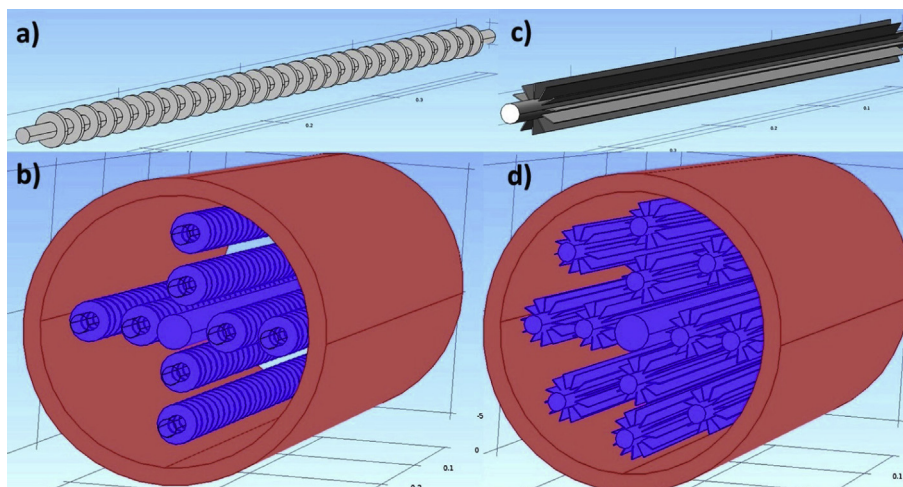


Fig. 3 – The geometries used for the study of scenario 2 and scenario 3. (a and c) shows a single heating/cooling equipped with transverse and longitudinal fins respectively. (b) shows the case of using 8 transverse finned tubes within the reactor and (d) the case of using 12 longitudinal finned tubes.

(LaNi₅ and the AB₂-alloy) the flatness factors and the hysteresis effects were measured experimentally.

Validation of numerical results

To validate the model, experiments were performed on a 0.8 g sample of LaNi₅ powder. The sample was synthesized by arc-melting and the phase purity was validated using XRD (Rietveld Analysis). The pressure-composition-isotherm (PCI) hydrogenation and dehydrogenation measurements were performed on a lab scale Sievert-type apparatus with a capacitance manometer (Druck PTX 620) at two different temperatures (25 and 50 °C). In addition, two different over pressures were used to demonstrate that the model could accommodate different driving pressures. Furthermore, at the same experimental apparatus, a larger amount of the AB₂-intermetallic (800 g) was used to validate the numerical model for the AB₂-intermetallic. Fig. 1(c) shows the hydrogenation capacity of such material, where the initial pressure at the beginning of the hydrogenation was 30 bar. Furthermore, Fig. 1(d) shows the temperature distribution for the hydrogenation process of the AB₂-intermetallic. The results of the modelling work compared to the experimental data show good agreement with a maximum deviation of less than 5% as illustrated in Fig. 1.

Tank design geometries

The metal hydride reactors considered in the current study were cylindrical, 316 stainless steel with a wall thickness of 3 mm. The total length of the reactor was 400 mm and the internal radius was 52.55 mm, resulting in an internal volume of $3.44 \cdot 10^{-3} \text{ m}^3$. For the first scenario the arrangements of the 18, 24, 30, 36, 48 and 60 embedded cooling/heating heat

exchangers, similar to those used by Anbarasu et al. [15], are presented in Fig. 2.

Fig. 3 illustrates scenarios 2 and 3 where the heat exchange tubes are equipped with Al extended surfaces. A single tube is shown in Fig. 3(a) and (c) equipped with transverse and longitudinal Al fins respectively. The tubes are shown assembled in the reactor for 8 transverse finned tube, Fig. 3(b) and 16 longitudinal Al fins, Fig. 3(d). In both geometries, the central tube enables the delivery of hydrogen along the axis of the bed.

Optimization objectives and control factors

The charging/discharging time is of major importance for the effective performance of a metal hydride tank. The purpose of the present optimization process is to obtain the parameters of the systems that lead to the minimum charging/discharging time (t_s). The charging/discharging time (t_s) is the time that is required for the average reacted fraction to rise from 0.1 to 0.9 for the hydrogenation process and to decrease from 0.9 to 0.1 for the dehydrogenation process. The optimization process was performed for the cases of the transverse and the longitudinal finned geometries and the optimized geometries will be compared to each other as well as with the embedded tube geometries. Table 2 presents the control factors and the values for the case of the transverse finned geometry and Table 3 presents the control factors and the values for the case of the longitudinal finned geometry.

Table 2 – Control factors for the transverse finned geometry and their values.

Parameters	Level				
	1	2	3	4	5
A: Number of fins (N)	20	25	30	35	40
B: Fin radius (R, mm)	5	6	6.5	7	8
C: Fin thickness (T, mm)	1	1.5	2	2.5	3.5

Table 3 – Control factors for the longitudinal finned geometry and their values.

Parameters	Level				
	1	2	3	4	5
A: Number of fins (N)	4	6	8	12	16
B: Fin length (L, mm)	2	3	3.5	4.5	5.5
C: Fin thickness (T, mm)	0.15	0.25	0.35	0.45	0.5

According to the optimization analysis, the optimum geometry for the transverse finned geometry is the combination of 30 fins with radius 8 mm and thickness 2 mm, while for the case of longitudinal fins the optimum geometry is the case with 12 fins with fin height 5 mm and thickness 0.5 mm. Fig. 4 shows the temperature distribution for both the transverse and the longitudinal case during the optimization process in two dimensions.

Results and discussion

Hydrogenation process

Scenario 1 – embedded cooling tubes

The initial temperature and the temperature of the cooling fluid during the hydrogenation process for all the studied cases was 20 °C and the supply pressure was selected at 15 bar, which is the pressure that a commercial electrolyser can provide [40]. The maximum time considered for the hydrogenation process was 2000 s, due to the strict requirements that a compression system demands. For scenario 1, it was observed that the average temperature of the hydride bed reaches the temperature of the cooling fluid faster for the cases where the reactor includes higher number of tubes; see Fig. 5(a). Due to the exothermic nature of the hydrogenation process the average bed temperature will initially increase. As the relatively large amount of heat is transferred away from the bed the average temperature will start to decrease towards that of the cooling fluid. The presence of 48 and 60 tubes provides the largest increase in primary heat transfer area

which in turns provides the most effective heat management, Fig. 5(a). For comparison, Fig. 5(b) shows the average bed temperature when using the combination of the heat exchangers and external jacket. In that case, the cooling rate is faster compared with the case of using only the cooling tubes and the system is able to reach the temperature of the cooling fluid faster. This temperature behaviour can impact the hydrogenation kinetics, as presented in Fig. 5(c) and (d). With the increase in the number of the cooling tubes in the system the hydrogenation process is faster compared to the cases with lower number of tubes. Fig. 5(e) also presents a comparison of cooling tubes only with the combination of both cooling tubes and external jacket for the different number of tubes with the time needed for the system to reach $X = 0.9$ as indicated in Results and Discussion section. As mentioned earlier, the maximum hydrogenation time considered was 2000 s, so this is the upper time shown in Fig. 5(e). According to Fig. 5(e), it would appear that the case of 18 and 24 tubes will fail to achieve the target hydrogen charging time of 2000 s where as for the case of 30 tubes either combination of tubes and tubes plus external jacket could reach $X = 0.9$ in less than 2000 s. For the case of 36 tubes, both cases can reach $X = 0.9$ in less than 2000 s, but the combination case (1726 s) is faster than the case of only cooling tubes (1970 s) at almost 4 min. When 48 tubes are inserted in such geometry for both cases the kinetics of the hydrogenation process are faster and the combination case is faster by more than 4 min. Finally, for the case of 60 tubes the combination of the cooling tubes with the external jacket has faster response for the hydrogenation process compared to the cooling tubes by almost 4 min. According to these results, it seems that the presence of 48 and 60 tubes in the metal hydride container can enhance the hydrogenation kinetics at least by 10 min, while the comparison between the usage of an extra external jacket seem to accelerate the reaction by almost 4 min for all the cases.

Scenario 2 – transverse finned tubes

According to the optimization study presented in Results and Discussion section, for the case of transverse fins, the optimization analysis, after simulating 125 different cases, the optimum geometry of the finned tubes is the combination of

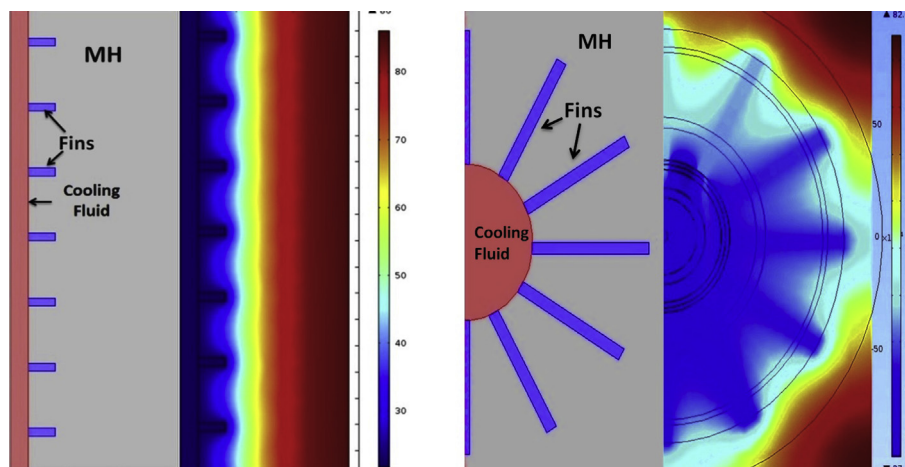


Fig. 4 – Temperature distribution for both the case of using the transverse and longitudinal fins during the optimization process.

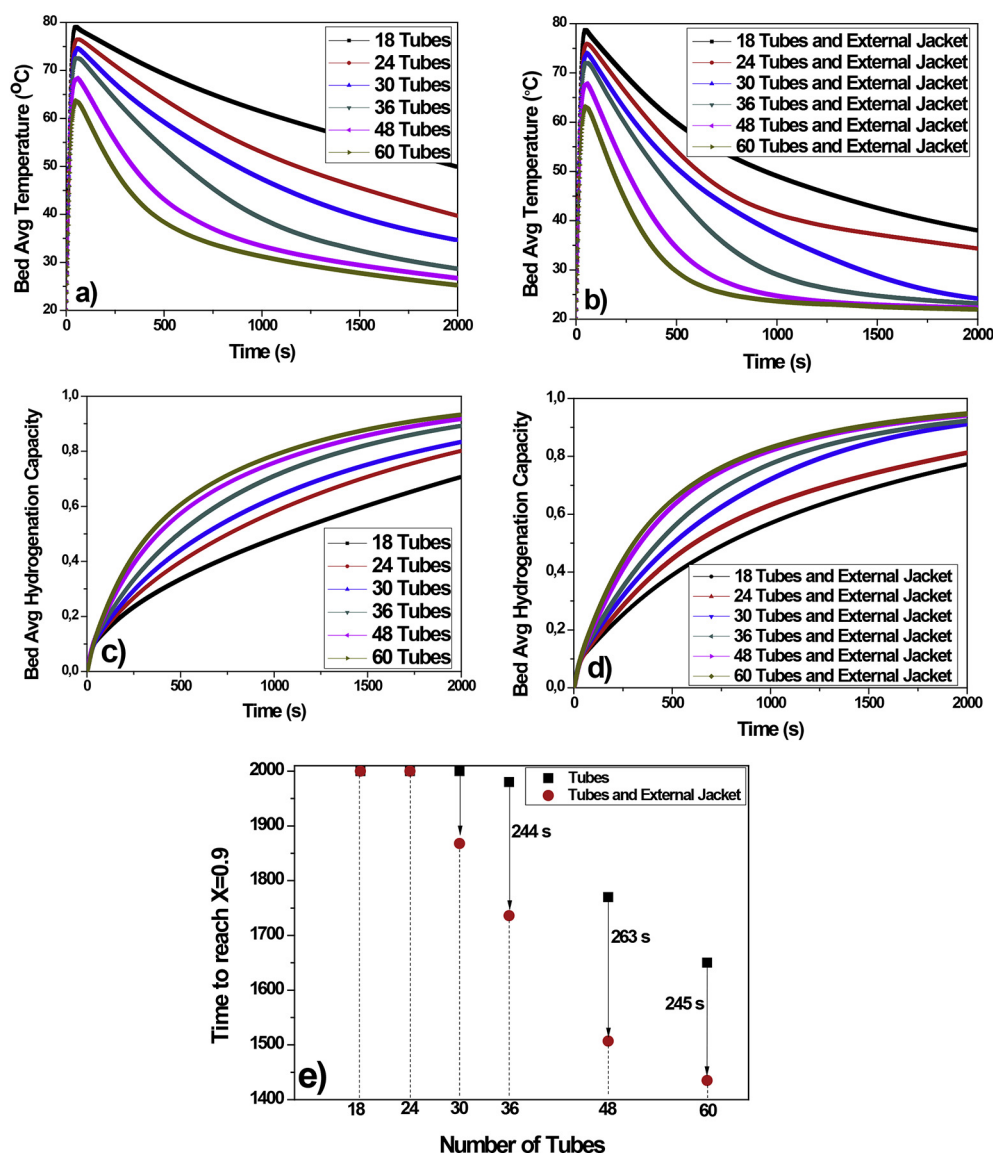


Fig. 5 – (a and b) shows the average temperature of the hydride bed during the hydrogenation process. (c and d) presents the average hydrogenation fraction during the hydrogenation process. (b and d) also presents the effect of the additional heat management when using an external jacket in combination with the embedded tubes. (e) shows the time needed for the bed to reach the average hydrogenation fraction $X = 0.9$ and compares the response time for the case of using only the tubes and the case of the combination of the tubes with the external jacket.

30 fins (distance between the fins 9.8 mm) with radius 8 mm and thickness 2 mm. Three different cases were examined in detail, 8–12 and 16 finned tubes, inserted inside the metal hydride container. The dimension and the total mass of the material inside the tank is the same as the previous scenario, in order to be able to compare the results for the current scenario with the previous one. Fig. 6(a) presents the kinetics of the hydrogenation behaviour of the material for the case of 8, 12 and 16 finned tubes when using only the finned tubes as the heat exchangers. Similar to the previous results, increasing of the number of the tubes can enhance the kinetics of the hydrogenation capacity due to the faster removal of the exothermically produced heat. Fig. 6(b) presents the kinetics of the hydrogenation behaviour of the material for the case of 8, 12 and 16 finned tubes when using the combination

of the finned tubes and an external jacket. According to the results, the combination of both jacket and tubes serve to increase the rate of the hydrogenation reaction. Fig. 6(c) shows the comparison of the time for the material to reach $X = 0.9$ during the hydrogenation process for 8, 12 and 16 finned tubes for finned tubes only and the combination of the finned tubes and the external jacket. In both cases the 8 finned tube arrangement is unable to reach the target of $X = 0.9$ in less than 2000 s. When using 12 finned tubes, it is only the combination of jacket and tubes that provide sufficient heat management that the system is able to reach $X = 0.9$ in under 2000 s. In fact $X = 0.9$ is reached in 1692 s, more than 5 min faster than that achieved by the finned tubes alone. Finally, when 16 finned tubes are installed in the container, in both cases the system is able to achieve $X = 0.9$ in less than 2000 s,

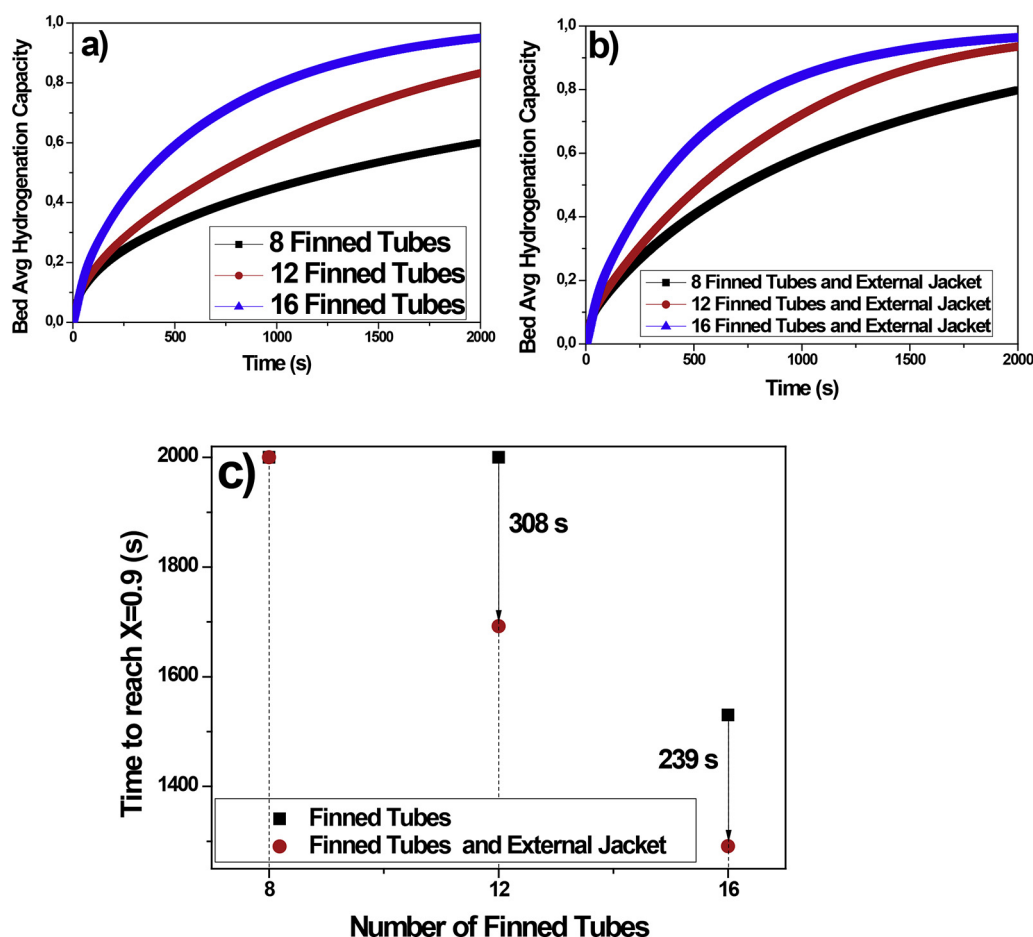


Fig. 6 – (a) The kinetics during the hydrogenation process for the case of using 8, 12 and 16 transverse finned tubes. **(b)** presents the case of using an additional external jacket in combination with the finned tubes. **(c)** shows the time needed for the hydride to reach the hydrogenation fraction $X = 0.9$ for both the cases with and without the external jacket combined with the finned tubes.

and the case of the combined heat management is almost 4 min faster.

Scenario 3 – longitudinal finned tubes

The current scenario involves the study of the usage of Al fins located longitudinal on the cooling tubes as presented in Fig. 3(c) and (d). According to the optimization study presented in Results and Discussion section, for the case of longitudinal fins, the optimization analysis, after simulating 125 different cases, the optimum geometry of the finned tubes is the case including 12 fins with fin height 5 mm and thickness 0.5 mm as presented in Fig. 3(c) and (d). Three different cases were examined in terms of the number of the finned tubes inserted inside the metal hydride container, again detailing 8, 12 and 16 finned tubes. The dimension and the total mass of the material inside the tank is the same as the previous scenario, in order to be able to compare the results for the current scenario with the previous scenarios discussed. The material inside the reactor has the properties of LaNi_5 as in the previous case. Fig. 7(a) presents the kinetics of the hydrogenation behaviour of the material for the case of 8, 12 and 16 finned tubes when using only the finned tubes as the heat exchangers. Similar to the results presented at the chapter which introduced the

transverse finned tubes, the increasing of the number of the tubes can enhance the kinetics of the hydrogenation capacity due to the faster removal of the produced heat during the hydrogenation process. Fig. 7(b) presents the kinetics of the hydrogenation behaviour of the material for the case of 8, 12 and 16 finned tubes when using the combination of the finned tubes and an external jacket. According to the results, the combination of both heat treatment ways increases the hydrogenation reaction. Fig. 7(c) shows the comparison of the time for the material to reach $X = 0.9$ during the hydrogenation process for 8, 12 and 16 finned tubes for both the case of using only the finned tubes and the combination of the finned tubes and the external jacket. When using 8 finned tubes, the system is not able to reach the target of $X = 0.9$ in less than 2000 s in both cases. When using 12 finned tubes, the case of the combined heat management is faster by slightly less than 4 min compared to the case of using only the finned tubes. For the case of using 16 finned tubes, the difference between the two heat management cases is small (around 36 s). According to these results, for the case of 16 finned tubes it seems that the presence of the external jacket does not affect the hydrogenation behaviour of the system and the presence of the finned tubes has a dominant effect on the performance of the

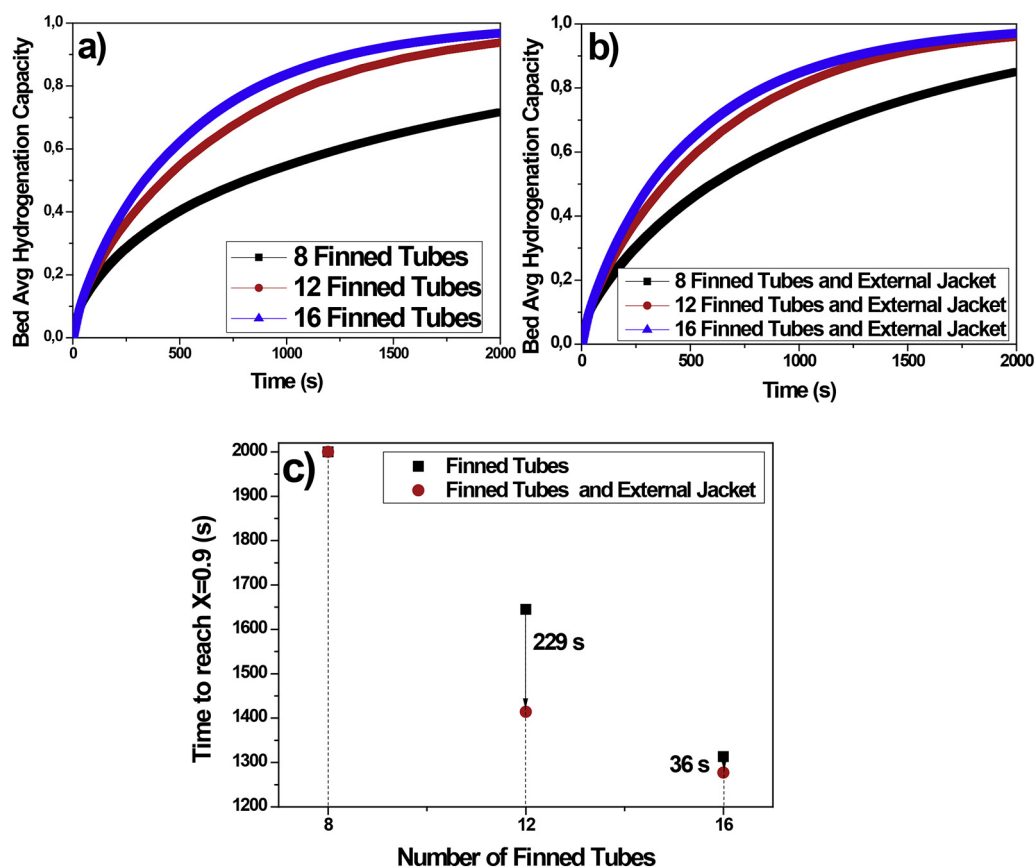


Fig. 7 – (a) The kinetics during the hydrogenation process for the case of using 8, 12 and 16 longitudinal finned tubes. (b) presents the case of using an additional external jacket in combination with the finned tubes. (c) shows the time needed for the hydride to reach the hydrogenation fraction $X = 0.9$ for both the cases with and without the external jacket combined with the finned tubes.

system. Furthermore, the kinetic behaviour showed that increasing the number of the tubes for the longitudinal finned case above 12 finned tubes has limited influence on the kinetics of the hydrogenation process.

Dehydrogenation process

Scenario 1 – embedded cooling tubes

According a previous work of the authors [37], the operation temperatures of the studied compression system were 20 °C (hydrogenation process) and 130 °C (dehydrogenation process), so in order to study the current system, the dehydrogenation process the initial temperature and the temperature of the heat exchange fluid is 130 °C. Fig. 8(a) shows the average temperature evolution within the metal hydride tank while the heat exchange system is the heating embedded tubes. The average hydride bed temperature initially reduces to its lowest value due to the rapid endothermic reaction that occurs during the dehydrogenation process and due to the low thermal conductivity of the hydride powder, the hydride uses the heat from itself to maintain the reaction while the temperature increases slowly towards the temperature of the heating fluid. As extracted from the results, the reactor with 48 and 60 tubes can recover the heat from the heating tubes faster than the other configurations. This is due to the heat transfer area aids

higher heat transfer rates from the heating tubes. The effect of adding an external jacket is presented in Fig. 8(b). As discussed in chapter regarding the embedded tubes the temperature recovery in that case is faster compared to the case when only the heat exchange is provided by the heating tubes. Fig. 8(c) and (d) shows the profile of the dehydrogenation capacity comparing the two different heat treatment cases. According to the results from Fig. 8(e), it seems that for the case of 18 and 24 tubes the hydrogen discharging time was 2000 s or more while for the case of 30 tubes, with the combination of heating fluid and the external jacket, it could reach $X = 0.1$ in less than 2000 s (at 1296 s). With 36 tubes, both cases can reach $X = 0.1$ in less than 2000 s, but once again the combination case (905 s) is faster than the case of only cooling tubes (1400 s) at almost 8 min. When 48 tubes are inserted in such geometry for both cases the kinetics of the hydrogenation process are faster and the combination case is faster by almost 8 min. Finally, for the case of 60 tubes the combination of the heating tubes with the external jacket has a faster response for the hydrogenation process compared to the cooling tubes by almost 7 min. These results indicate that the presence of 48 and 60 tubes in the metal hydride container can enhance the dehydrogenation kinetics at least by 10 min, while the addition of an extra external jacket accelerates the reaction by almost 7–8 min for all the cases. It has to be noted that by comparing the results

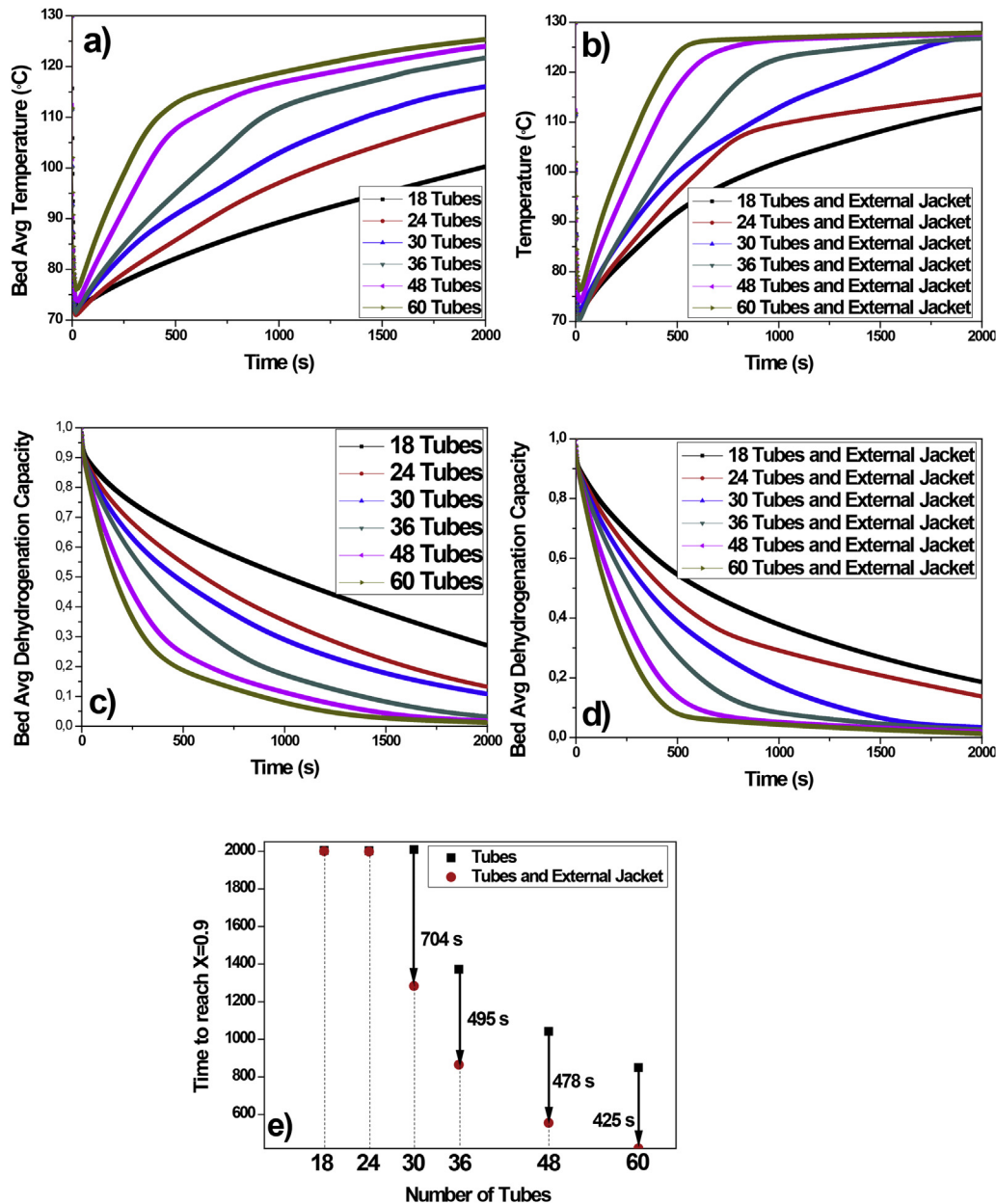


Fig. 8 – (a and b) shows the average temperature of the hydride bed during the dehydrogenation process. (c and d) presents the average dehydrogenation fraction during the dehydrogenation process. (b and d) also presents the effect of the additional heat management when using an external jacket in combination with the embedded tubes. (e) shows the time needed for the bed to reach the average dehydrogenation fraction $X = 0.1$ and compares the response time for the case of using only the tubes and that of the case of combining the tubes with the external jacket.

extracted from the hydrogenation process and the dehydrogenation process, the external jacket has more impact at the dehydrogenation process rather than the hydrogenation.

Scenario 2 – transverse finned tubes

For this study the geometry used is exactly the same with the geometry described to the chapter regarding the transverse finned tubes. Three different cases were examined in terms of the number of the finned tubes inserted inside the metal hydride container, 8, 12 and 16 finned tubes. Fig. 9(a) presents the kinetics of the dehydrogenation behaviour of the material for the case of 8, 12 and 16 finned tubes when using only the

finned tubes as the heat exchangers. Similar to the previous results, increasing of the number of the tubes can enhance the kinetics of the dehydrogenation capacity due to the faster heat supply from the heating tubes which results to the higher heat exchange area between the tubes and the hydride. Fig. 9(b) presents the kinetics of the dehydrogenation behaviour of the material for the case of 8, 12 and 16 finned tubes when using the combination of the finned tubes and an external jacket. The combination of both heat treatment ways increases the kinetics of the hydrogenation reaction. Fig. 9(c) shows the comparison of the time for the material to reach $X = 0.1$ during the dehydrogenation process for 8, 12 and 16 finned tubes for

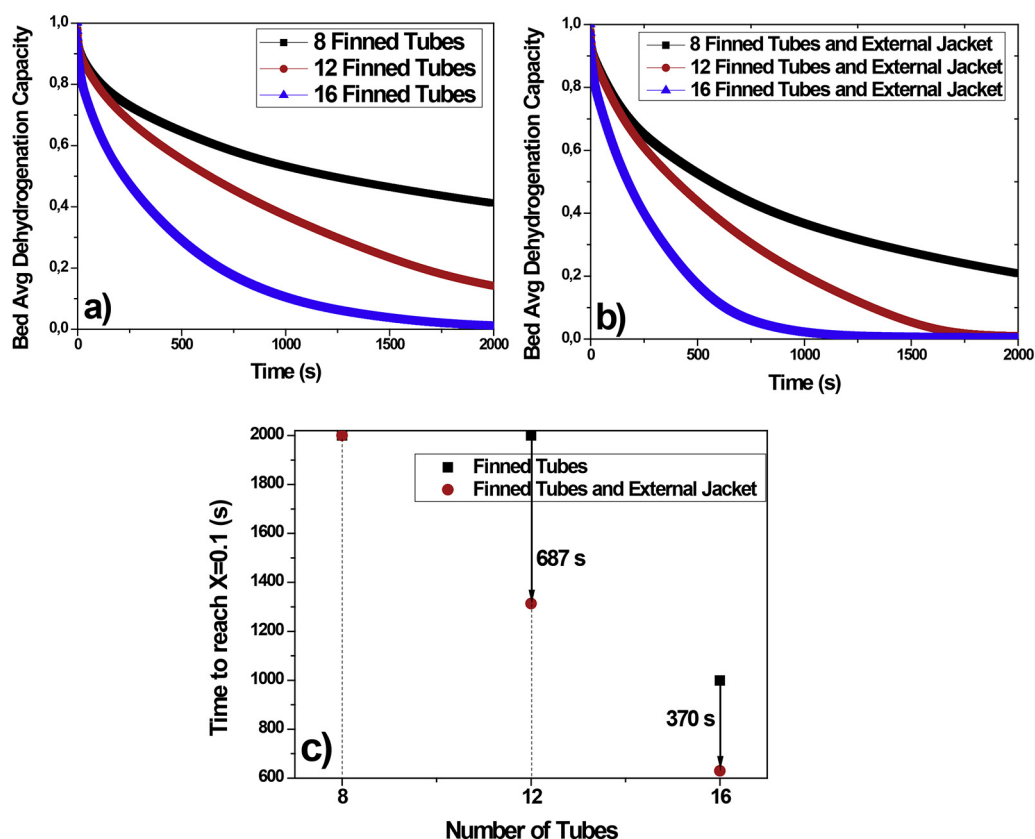


Fig. 9 – (a) The kinetics during the dehydrogenation process for the case of using 8, 12 and 16 transverse finned tubes. (b) presents the case of using an additional external jacket in combination with the finned tubes. (c) shows the time needed for the hydride to reach the hydrogenation fraction $X = 0.1$ with and without the external jacket combined with the finned tubes.

the finned tubes and the combination of the finned tubes and the external jacket. When using 8 finned tubes, the system is not able to reach the target of $X = 0.1$ in less than 2000 s in both cases. When using 12 finned tubes only the system is not able to reach the target capacity at 2000 s, while for the combination heat management case the system is able to reach $X = 0.1$ in 1313 s, more than 11 min faster. Finally, when 16 finned tubes are installed in the metal hydride tank, in both cases the system is able to achieve $X = 0.1$ in less than 2000 s. The case of the combination of both finned tubes and external heating is more than 6 min faster compared to the case of using only the finned tubes.

Scenario 3 – longitudinal finned tubes

For this scenario, the Al fins are located longitudinal on the cooling tubes as presented in Fig. 3(c) and (d). Three different cases were examined in terms of the number of the finned tubes inserted inside the metal hydride container. That is for 8, 12 and 16 finned tubes. Fig. 10(a) shows the kinetics of the dehydrogenation behaviour of the material for the case of 8, 12 and 16 finned tubes when using only the finned tubes as the heat exchangers, while Fig. 10(b) presents the kinetics of the dehydrogenation behaviour of the material for the case of 8, 12 and 16 finned tubes when using the combination of the finned tubes and an external jacket. According to the results, the combination of both heat treatment ways increases the

kinetics of the hydrogenation reaction. Fig. 10(c) shows the comparison of the time for the material to reach $X = 0.1$ during the dehydrogenation process for 8, 12 and 16 finned tubes for both the case of using only the finned tubes and the combination of the finned tubes and the external jacket. When using 8 finned tubes, the system is not able to reach the target of $X = 0.1$ in less than 2000 s in both cases as indicated to the previous scenarios. When using 12 finned tubes, in both cases the dehydrogenation process becomes faster and both cases are able to achieve $X = 0.1$ in less than 2000 s. The single tube case can deliver the dehydrogenation process at 1147 s, while the combination of tubes and the external jacket is faster by almost 5 min and can deliver $X = 0.1$ in 849 s. Finally, when 16 finned tubes are installed in the metal hydride tank, in both cases the system is able to achieve $X = 0.1$ in less than 2000 s. The case of the combination of both finned tubes and external heating is almost 3 min faster compared to the case of using only the finned tubes within the tank.

Comparison of the LaNi_5 and AB_2 -type materials performance during the hydrogenation process

The performance of a metal hydride hydrogen compression system depends on the characteristics of the operating hydride materials. The most important characteristics are the charging/discharging times (kinetics), the maximum

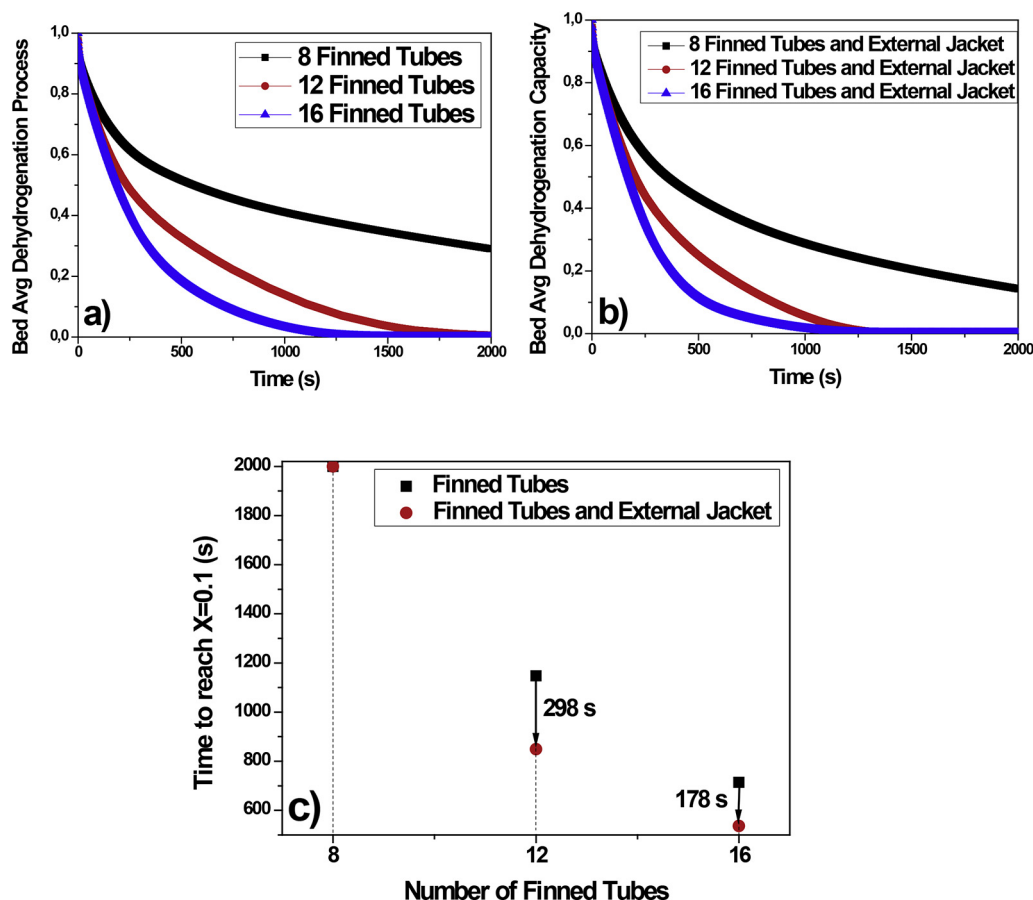


Fig. 10 – (a) The kinetics during the hydrogenation process for the case of using 8, 12 and 16 longitudinal finned tubes. (b) presents the case of using an additional external jacket in combination with the finned tubes. (c) shows the time needed for the hydride to reach the hydrogenation fraction $X = 0.1$ for both the cases with and without the external jacket combined with the finned tubes.

reversible amount of hydrogen that the material can store/release and the operating temperatures. LaNi_5 is a well-studied material with relatively fast kinetics, a reversible capacity of 1.45 wt%, low operating temperatures and with a flat plateau and small hysteresis. In comparison the Laves-phase AB_2 -type intermetallics also operates at low temperatures, can achieve higher dehydrogenation pressures but has a sloping plateau with higher hysteresis effects. Usually, the maximum hydrogenation capacity of such materials is in the range of 1.2–2.5 wt.%. Both materials are possible candidates for the first stage of a two-stage compression system as discussed in Ref. [36]. The first material is LaNi_5 and the second material is a Ti–Zr–Mn based AB_2 -type intermetallic synthesized, characterized and tested at the University of Nottingham. The materials performance was compared for all the geometries described for scenarios 1, 2 and 3. Fig. 11(a) presents the hydrogenation kinetics for the LaNi_5 and the AB_2 -type for the case of the container including cooling tubes. According to the results from the previous chapter regarding the embedded tubes, the case of 48 and 60 embedded tubes presents the fastest kinetics and for clarity Fig. 11(a) deals only with these two cases. As extracted from Fig. 11(a), the AB_2 -type presents faster kinetics at the beginning of the hydrogenation process compared to LaNi_5 , but afterwards LaNi_5

reaches $X = 0.9$ faster by less than 1 min for the cases of using 36 and 60 tubes and 71 s for the case of using 48 tubes as shown in Fig. 11(b). For that case it seems that the kinetic performance of both materials is almost the same. Fig. 11(c) and (d) shows the comparison for the two materials for the case of using the transverse finned tubes. For the case of using 8 and 12 finned tubes both materials are unable to achieve $X = 0.9$ in less than 2000 s and for the case of using 16 finned tubes, both materials can achieve $X = 0.9$ in less than 2000 s and the AB_2 -type can be faster than LaNi_5 by 65 s. Fig. 11(e) and (f) shows the comparison for the two materials for the case of using the longitudinal finned tubes. For the case of using 8 finned tubes both materials are unable to achieve $X = 0.9$ in less than 2000 s and for the case of using 12 and 16 finned tubes, both materials can achieve $X = 0.9$ in less than 2000 s. Specifically, for the case of using 12 finned tubes LaNi_5 can achieve $X = 0.9$ in 1600 s while the AB_2 -type can achieve $X = 0.9$ faster by more than 4 min and by using 16 finned tubes LaNi_5 can achieve $X = 0.9$ in 1270 s while the AB_2 -type can achieve $X = 0.9$ faster by more than 5 min. A very interesting outcome of the current analysis is the fact that for the embedded tube and the transverse finned tube geometry, there is not much difference in the time both materials can achieve $X = 0.9$ (the maximum difference is about 1 min for

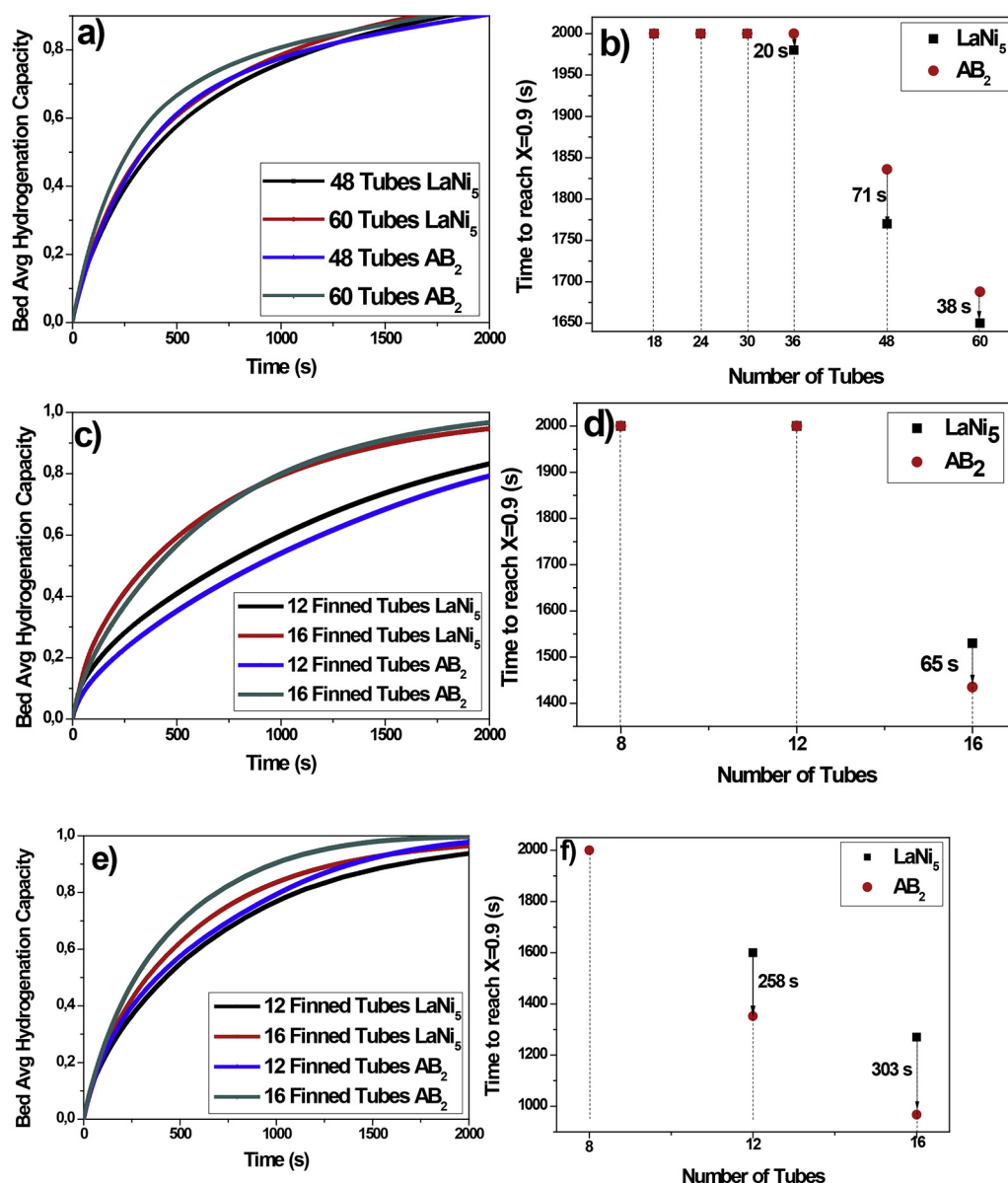


Fig. 11 – Comparison of the two candidate metal hydride materials proposed for the first stage of a compression system, LaNi₅ and an AB₂-type intermetallic. (a and b) presents the comparison of the performance of the two materials when using the embedded tubes as the heat management agents. (c and d) shows the performance and comparison of the two materials for the case of using transverse finned tubes, while (e and f) shows the comparison when using longitudinal finned tubes.

both materials), while for the case of the longitudinal finned geometry, the AB₂-type material seems to present a much faster response comparing to LaNi₅.

Comparison of scenarios 1, 2 and 3

For the three different heat management scenarios, the most effective heat management measures were: 48 embedded heat exchanger tubes, 60 embedded heat exchanger tubes, 12 transverse finned tubes, 16 transverse finned tubes, 12 longitudinal finned tubes and 16 longitudinal finned tubes. Fig. 12(a) shows the comparison of all six cases, when using LaNi₅ as the operating material. It has to be noted that all the initial conditions of temperature and pressure are the same

for the comparison study. For clarity, Fig. 12(b) shows the same comparison but focused on the last 1000 s of the hydrogenation reaction. According to the results, when using LaNi₅ as the hydride material the fastest kinetics are observed in the case of using 16 longitudinal finned tubes as the heat exchangers, followed by the case of using 16 transverse finned tubes (almost 3 min slower), 12 longitudinal finned tubes (almost 4.5 min slower) and 60 embedded cooling tubes (almost 4.5 min slower). The cases of using 48 embedded cooling tubes and 12 transverse finned tubes are slower and the case of using 12 transverse finned tubes is not able to reach $X = 0.9$ in less than 2000 s. During the first 1000 s of the hydrogenation reaction most of the heat treatment cases present almost the same kinetic behaviour, especially for the

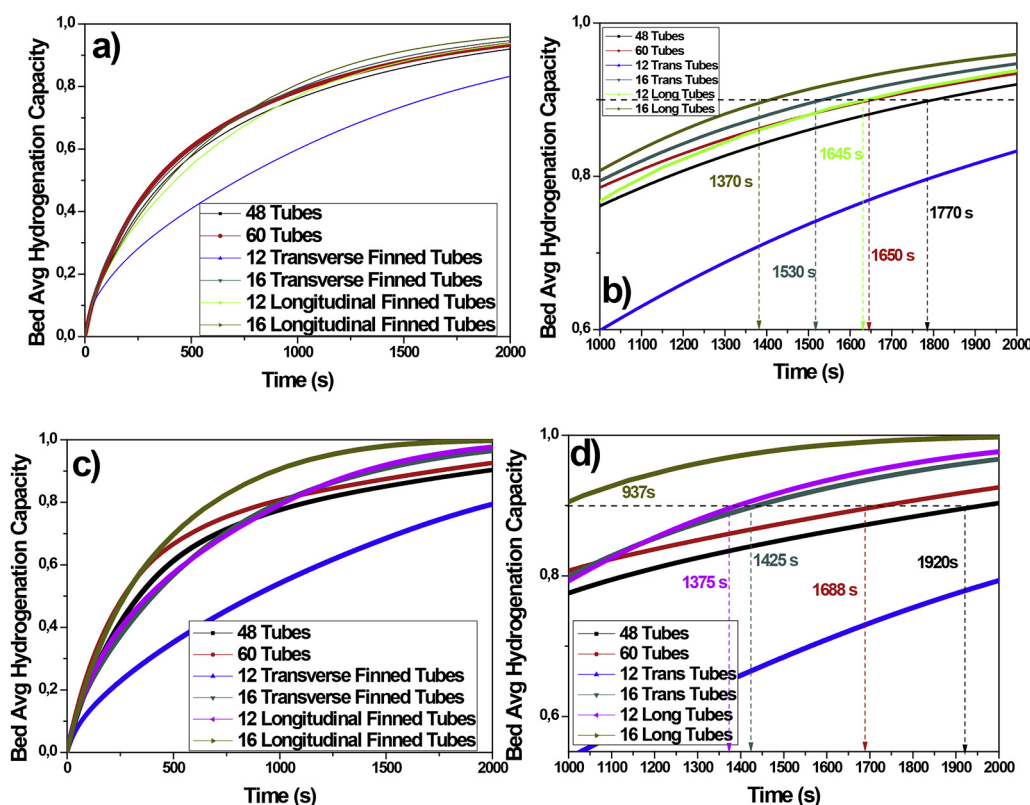


Fig. 12 – Comparison of all the studied heat management scenarios. (a and b) presents the comparison when using LaNi_5 as the operating material. (c and d) shows the comparison when using the AB_2 -type intermetallic as the operating material.

cases of using 60 embedded tubes, 16 transverse finned tubes and 16 longitudinal finned tubes. Fig. 12(c) shows the comparison of the heat management cases when using the AB_2 -type material described earlier as the operating hydride. In that case, all the conditions were exactly the same with the conditions used for the case of using LaNi_5 as the operating material. For the first 1000 s, according to Fig. 12(c) it is found that, unlike the case of using LaNi_5 , for the AB_2 -type, the heat management cases present different kinetic behaviour, where the 16 longitudinal finned tubes and the 60 embedded tubes present fast kinetics at the beginning of the hydrogenation reaction, while all the other cases present a slower response. For the final 1000 s of the hydrogenation reaction, as illustrated in Fig. 12(d), the vessel with the 16 longitudinal finned tubes can reach $x = 0.9$ in less than 1000 s (937 s), followed by the case of the vessel with 12 longitudinal finned tubes (slower by 7.5 min) and the vessel with 16 transverse finned tubes (slower by more than 8 min). The vessel equipped with 60 embedded cooling tubes delivers fast kinetics at the beginning of the reaction, reaches a smoother reaction rate at the final stage of the reaction and is able to reach $X = 0.9$ in more than 12 min, compared with the case of the vessel equipped with 16 longitudinal finned tubes.

As shown from the above results, it is clear that the case of using a vessel equipped with 16 longitudinal finned tubes is the most efficient way to enhance the hydrogenation kinetics when using both LaNi_5 and the AB_2 -alloy. When using LaNi_5 as the operating hydride the case of the vessel equipped with 60 embedded cooling tubes presents the same kinetic behaviour

with the case of the vessel equipped with 12 longitudinal finned tubes, so in that way, by using extended surfaces to enhance the heat exchange can reduce the total number of tubes from 60 to 12. For the case of using the AB_2 -type material as the operating hydride the performance of the extended surfaces is more dominant and more effective comparing to the case of using the embedded tubes, especially for the case of the longitudinal extended surfaces.

Conclusions

In the current work, three different heat management scenarios were examined, for both the hydrogenation and dehydrogenation process. The three different scenarios included the case of the embedded cooling/heating tubes, the case of using transverse finned tubes and the case of the longitudinal finned tubes. A detailed optimization study was also presented in order to choose the optimized geometries for the finned tube geometries and the optimum geometry for the transverse finned tubes is the combination of 30 fins (distance between the fins 9.8 mm) with radius 8 mm and thickness 2 mm and the optimum geometry for the longitudinal finned tubes is the combination of 12 fins with each fin height 5 mm and thickness 0.5 mm. Furthermore, two different types of hydrides were studied as possible candidate materials for using as the first stage alloys in a two-stage metal hydride hydrogen compression system, in terms of the hydrogenation kinetics under the same conditions and the same heat

management scenarios. The results showed that for the hydrogenation process, when using LaNi_5 for the case in vessel equipped with the embedded tubes the cases of 48 and 60 tubes are able to achieve $X = 0.9$ in less than 2000 s, while the use of an additional external jacket can enhance the hydrogenation process by almost 4 min, for the cases of 36, 48 and 60 tubes. For the dehydrogenation process the results showed that the usage of 36, 48 and 60 tubes can ensure that the system is able to reach $X = 0.1$ in less than 2000 s, while the use of the additional external jacket can accelerate the reaction by almost 7 min. For a vessel equipped with transverse finned tubes, for the hydrogenation process, the results showed that only the case of 16 finned tubes can reach $X = 0.9$ within 2000 s, but when using an additional jacket both the cases of 12 and 16 finned tubes can achieve the target. Furthermore, the presence of the external jacket accelerates the process by 4–5 min. For the dehydrogenation process the results showed that only the case of 16 finned tubes can reach $X = 0.1$ within 2000 s, and the presence of the external jacket allows the system to achieve $X = 0.1$ in less than 2000 s for both the cases of 12 and 16 finned tubes. For a vessel equipped with longitudinal finned tubes the hydrogenation reaction is fast and both the cases of the vessel equipped with 12 and 16 finned tubes. They both can achieve $X = 0.9$ in less than 2000 s and the presence of the additional external jacket does not affect the reaction in a great manner for the case of 16 finned tubes. For the dehydrogenation process the case of both 12 and 16 finned tubes allows the system to achieve $X = 0.1$ in less than 2000 s. Finally, for the comparison of the two candidate materials for the first stage of the metal hydride hydrogen compression system, LaNi_5 and the AB_2 -type (Ti–Zr–Mn based) intermetallic showed that the hydrogenation reaction kinetics in all the heat management scenarios had generally similar behaviour. However, while the case of the embedded tubes and the transverse finned tubes is almost the same, for the case of the longitudinal fins the AB_2 -type intermetallic presents a much faster response than LaNi_5 (4.5 min for the 12 finned tubes and 5 min for the 16 finned tubes), making the AB_2 -type (Ti–Zr–Mn based) intermetallic a very potential candidate for the first stage material of the two-stage compression system.

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Nomenclature

C_a	Absorption Reaction Constant, s^{-1}
C_d	Desorption Reaction Constant, s^{-1}
C_p	Specific Heat, $\text{J/kg}\cdot\text{K}$
E_a	Activation Energy for Absorption, J/molH_2
E_d	Activation Energy for Desorption, J/molH_2
h	Heat Transfer Coefficient, $\text{W/m}^2\text{K}$
k	Thermal Conductivity, $\text{W/m}\cdot\text{K}$
K	Permeability, m^2

M	Molecular Weight, kg/mol
m	Kinetic Expression
n	Number of Hydrogen Moles
P	Pressure, bar
R	Gas Global Constant, $\text{J/mol}\cdot\text{K}$
t	Time, s
T	Temperature, K
v	Gas Velocity, m/s
V	Volume, m^3

Subscripts

a	Absorption
d	Desorption
e	Effective
eq	Equilibrium
f	External Heater/Cooler
g	Gas
i	Initial
s	Solid
ss	Saturation

Greek letters

ε	Porosity
μ	Dynamic Viscosity, kg/ms
ρ	Density, kg/m^3
ΔH	Reaction Enthalpy, J/mol
ΔS	Reaction Entropy, $\text{J/mol}\cdot\text{K}$

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